

Daily Number Theory Drill #4

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Prime Factor Algebra · Difference of Squares · Bounding Factors.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Prime factors, divisibility rules, and quick remainders.

- Find the largest prime factor of 1001.
- What is the last digit of 2^{2026} ?
- What is the remainder when $2^{10} + 3^{10}$ is divided by 5?
- Find the greatest common divisor of 120 and 84.
- How many positive integer factors does 144 have?
- What is the remainder when $11 \times 12 \times 13$ is divided by 10?
- What is the smallest positive integer divisible by 2, 3, 4, 5, and 6?
- Find the sum of the prime factors of 210.
- What is the remainder when $10^{2026} - 1$ is divided by 9?
- Find the value of k if 2^k is the highest power of 2 dividing $20!$.

Section B Manipulation Drills

(≤50 s each)

Modulo arithmetic properties, CRT basics, and divisor counting.

- Find the smallest positive integer n such that $15n$ is a perfect square.
- Find all single digits d such that the four-digit number $5d72$ is divisible by 9.
- Find the number of trailing zeros in $50!$.
- What is the remainder when 5^{2025} is divided by 7?

5. If $x \equiv 3 \pmod{7}$, find the remainder when $x^2 + 4x + 5$ is divided by 7.
6. Find the smallest positive integer that leaves a remainder of 2 when divided by 3, and 3 when divided by 5.
7. Find the sum of all positive integer divisors of 100.
8. What is the last digit of $3^{101} + 7^{101}$?
9. Find the number of positive divisors of $2^3 \times 3^4 \times 5$.
10. Find the largest three-digit number that is a multiple of 11 and 13.

Section C Substitution & Structure(≤ 90 s each)

Look for algebraic structures like difference of squares and combinatorial counting strategies.

1. Find all pairs of positive integers (x, y) such that $x^2 - y^2 = 45$. (after SMC 2018 Q15)
2. The prime number p can be expressed as $p = a^2 - b^2$, where a and b are also prime numbers. Find the value of p . (after SMC 2011 Q19)
3. Find the largest integer n such that $n^3 + 100$ is divisible by $n + 10$. (after MAT 2016 Q1H)
4. How many ordered pairs of positive integers (x, y) satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$? (after SMC 2015 Q21)
5. Find all integer values of x for which $\frac{x^2+5x+14}{x+3}$ is an integer. (after SMC 2019 Q20)
6. How many divisors of 10^{10} are perfect squares? (after SMC 2021 Q22)
7. How many positive integers $n \leq 1000$ are not divisible by 2, 3, or 5? (after SMC 2017 Q19)
8. Given that p and q are prime numbers satisfying $p = q^2 - 36$, find the value of p . (after SMC 2014 Q22)

Section D Challenge

(BMO1 difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find all pairs of positive integers (m, n) such that $3^m - 2^n = 1$. (after BMO1 2008 Q1)
2. For how many positive integers n is $n^4 + 4^n$ a prime number? (after BMO1 2013 Q2)
3. A sequence is defined by $a_1 = 1$, $a_2 = 1$, and $a_{n+2} = a_{n+1} + a_n$. What is the remainder when a_{2026} is divided by 3? (after SMC 2022 Q25)
4. Let p, q , and r be prime numbers such that $p = 9q^2 - r^2$. Find the value of p . (after BMO1 2010 Q1)

5. Find the largest positive integer n such that 2^n divides $3^{1024} - 1$. (*after BMO1 2019 Q3*)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths